4 a

ESCRIBIR UNA ECUACIÓN LINEAL HOMOGÉNEA CON COEFICIENTES CONSTANTES

$$y'' + P y' + (q_0 + q_1 \lambda) y = 0$$

COMO UNA ECUACIÓN DE STURM LIOUVILLE.
ESCRIBIR LA PROPIEDAD DE ORTOGONALIDAD QUE CUMPLEN LAS AUTOFUNCIONES SI IMPONEMOS CONDICIONES DE CONTORNO HOMOGÉNEAS Y SEPARADAS

UNA ECUACION DE STURM - LIOUVILLE TIENE LA FORMA

$$\frac{d}{dx}\left(p(x) \frac{dY(x)}{dx}\right) + \left[q(x) + \lambda r(x)\right] Y(x) = 0$$

$$\Downarrow$$

$$p' Y' + p Y'' + (q + \lambda r) Y = 0$$

$$\Downarrow$$

$$y'' + \frac{p'}{p} y' + \frac{q + \lambda r}{p} y = 0$$

FORMA CANÓNICA DE ST-L.

$$\Downarrow$$

$$\frac{p'}{p} = P = \text{CONST} \Rightarrow \frac{dp}{p} = dx\, P$$

$$\log p = P x + \text{CONST.}$$

$$\boxed{p(x) = C \exp(Px)}$$

$$\frac{q + \lambda r}{p} = q_0 + \lambda q_1 = \text{CONST}$$

$$\Downarrow$$

$$\boxed{q(x) = C\, q_0 \exp(Px)}$$

$$\boxed{r(x) = C\, q_1 \exp(Px)}$$

TEOREMA 3.1 DE LOS APUNTES

EL OPERADOR DIFERENCIAL $\mathcal{L} = -\left(\frac{d}{dx}\left(p(x)\frac{d}{dx}\right) + q(x)\right)$

ES AUTOADJUNTO PORQUE $p(x), q(x)$ SON FUNCIONES REALES, Y ES HERMÍTICO CUANDO LAS CONDICIONES DE CONTORNO SON <u>HOMOGÉNEAS</u> Y <u>SEPARADAS</u>

$$A_1 y(a) + A_2 y'(a) = 0 = B_1 y(b) + B_2 y'(b)$$

$$\left((f, \mathcal{L}g)_1 = \int_a^b dx\, f^* (\mathcal{L}g) = \int_a^b dx\, (\mathcal{L}f)^* g = (\mathcal{L}f, g)_1 \right.$$

$\Rightarrow$ NO TÉRMINOS DE FRONTERA $\Big)$

TEOREMA 3.2 $\quad r > 0$ EN $[a, b]$

$$\Updownarrow$$

$$C q_1 > 0 \quad \leftarrow \text{EN ESTE CASO}$$

$$\Downarrow$$

AUTOVALORES $\in \mathbb{R}$

TEOREMA 3.3 $\Rightarrow$ SI $\mathcal{L}$ ES HERMÍTICO

y y_λ SON AUTOFUNCIONES REALES CON AUTOVALORES REALES

$$\boxed{(y_\lambda, y_\mu)_r = \int_a^b dx\, r(x)\, y_\lambda(x)\, y_\mu(x) = 0}$$

SI $\lambda \neq \mu$

DONDE AHORA $r(x) = C q_1 e^{px}$

(b) ESCRIBIR ECUACIÓN DE EULER HOMOGÉNEA

$$x^2 y'' + A x y' + (b_0 + b_1 \lambda) y = 0$$

COMO EC. DE STURM-LIOUVILLE. + PROP. ORT. ETC. COMO ANTES

$$\frac{d}{dx}\left(p(x) \frac{dy(x)}{dx}\right) + \left[q(x) + \lambda r(x)\right] y(x) = 0$$

$$\Downarrow$$

$$p' y' + p y'' + (q + \lambda r) y = 0$$

$$\Downarrow$$

$$y'' + \frac{p'}{p} y' + \frac{q + \lambda r}{p} y = 0$$

$$\frac{p'}{p} = \frac{A}{x} \Rightarrow \frac{dp}{p} = A \frac{dx}{x} \quad \log p = A \log x + \text{CONST}$$

$$\boxed{p = C x^A}$$

$$\frac{q+\lambda r}{p} = \frac{b_0 + b_1 \lambda}{x^2}$$

$$\boxed{q = c\, b_0\, x^{A-2}} \qquad \boxed{r = c\, b_1\, x^{A-2}}$$

EN LAS HIPÓTESIS DE T. 3.1, 3.2, 3.3

RELAC. DE ORTOGONALIDAD ES

$$\boxed{\int_a^b dx \; x^{A-2} \; y_\lambda(x) \; y_\mu(x) = 0 \quad \text{SI } \lambda \neq \mu}$$

5

ESCRIBIR EC. DE BESSEL $\big(y(x) \big)$

$$x^2 y'' + x\, y' + (x^2 - p^2)\, y = 0$$

AUTOVALOR $\sim \mu^2$

COMO EC. DE STURM - LIOUVILLE.

$$y'' + \frac{p'}{p} y' + \frac{q + \lambda r}{p} y = 0$$

$$\frac{p'}{p} = \frac{1}{x} \Rightarrow \frac{dp}{p} = \frac{dx}{x}$$ (COMO ANTES, $A = 1$)

$$\boxed{p(x) = C x}$$

$$\boxed{q(x) = C x}$$

$$\boxed{r(x) = \frac{C}{x}}$$

No

6 CONSIDERAR EL PROBLEMA STURM - LIOUVILLE

$$\frac{d}{dx}\left(e^{2x}\frac{dy}{dx}\right)+\lambda e^{2x}y(x)=0$$

PROBLEMA TRAMPA: FACIL

EN INTERVALO $[0,1]$ CON CONDICIONES CONTORNO

$$y(0)=0 \qquad y(1)=0$$

(SEPARADAS Y HOMOGENEAS)

a DETERMINAR AUTOVALORES Y AUTOFUNCIONES.

EXPANDIMOS EC.

$$e^{2x}y''+2e^{2x}y'+\lambda e^{2x}y=0$$

$$\Downarrow$$

$$\boxed{y''+2y'+\lambda y=0} \Rightarrow$$ EC. DIFERENCIAL (LINEAL, II ORD) HOMOGÉNEA CON COEFICIENTES CONSTANTES.

BUSCAMOS SOLUCIONES DEL TIPO

$$y=e^{mx}$$

$$(m^2+2m+\lambda)e^{mx}=0$$

$$e^{mx}>0 \quad \text{EN } [0,1]$$

$$\Downarrow$$

$$\boxed{m^2+2m+\lambda=0} \qquad \boxed{m=-1\pm\sqrt{1-\lambda}}$$

I $\lambda < 1$ $\quad y(x) = C_+ e^{(-1+\sqrt{1-\lambda})x} + C_- e^{(-1-\sqrt{1-\lambda})x}$

$$0 = y(0) = C_+ + C_- \Rightarrow C_- = -C_+$$

$$0 = y(1) = C_+ e^{(-1+\sqrt{1-\lambda})} + C_- e^{(-1-\sqrt{1-\lambda})}$$

$$= C_+ \left(e^{-1+\sqrt{1-\lambda}} - e^{-1-\sqrt{1-\lambda}} \right)$$

$$-1 + \sqrt{1-\lambda} \overset{?}{=} -1 - \sqrt{1-\lambda} \Rightarrow \lambda = 1$$ NO ESTÁ PERMITIDO

$$\Rightarrow C_+ = C_- = 0$$

$\Rightarrow$ SOLAMENTE SOLUCIÓN TRIVIAL

II $\lambda = 1$ $\Rightarrow$ $m = -1$ EL MÉTODO NOS DA 1 SOLUCIÓN

MÉTODO DE LIOUVILLE COMO EN APUNTES

$$y = (c_1 + c_2 x) e^{mx}$$

PORQUE SI $y = x e^{mx}$

$$y' = e^{mx} + m x e^{mx} \qquad y'' = 2m e^{mx} + m^2 x e^{mx}$$

$$y'' + 2y' + \lambda y = 2m e^{mx} + m^2 x e^{mx} + 2e^{mx} + 2mx e^{mx} + \lambda x e^{mx}$$

$$= \underbrace{(2m+2)}_{0} e^{mx} + \underbrace{(m^2 + 2m + \lambda)}_{0} x e^{mx}$$

CONDICIONES DE CONTORNO

$0 = y(0) = C_1 \qquad 0 = y(1) = C_2 e^{-1}$

$\Rightarrow$ SOLAMENTE SOLUCIÓN TRIVIAL

(III) $\lambda > 1$ NOS VA A DAR SOLUCIONES

$$y(x) = C_+ e^{(-1+i\sqrt{\lambda-1})x} + C_- e^{(-1-i\sqrt{\lambda-1})x}$$

$$= e^{-x}\left(C_+ e^{i\sqrt{\lambda-1}\,x} + C_- e^{-i\sqrt{\lambda-1}\,x}\right)$$

$$\left(= e^{-x}\left[(C_+ + C_-)\cos(\sqrt{\lambda-1}\,x) + i(C_+ - C_-)\sin(\sqrt{\lambda-1}\,x)\right]\right)$$

$0 = y(0) = C_+ + C_- \Rightarrow C_- = -C_+$

$0 = y(1) = e^{-1} C_+ \left(e^{i\sqrt{\lambda-1}} - e^{-i\sqrt{\lambda-1}}\right)$

$\Rightarrow C_+ \sin\sqrt{\lambda-1} = 0$

PARA EVITAR LA SOLUCIÓN TRIVIAL $C_+ = C_- = 0$
EXIGIMOS

$$\boxed{\sin\sqrt{\lambda-1} = 0}$$

$$\Downarrow$$

$$\sqrt{\lambda-1} = n\pi \quad , \quad n \in \mathbb{N}^+$$

$$\boxed{\lambda = 1 + (n\pi)^2} \quad n \in \mathbb{N}^+$$

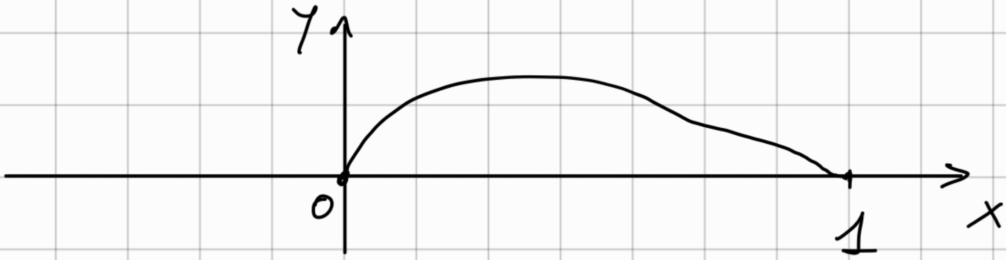
NO $\lambda = 1$

AUTOFUNCIONES $\quad y(x) = e^{-x} C \sin(n\pi x), \; n \in \mathbb{N}^+$

(b) DIBUJAR PRIMERAS 3 AUTOFUNCIONES Y ESPECIFICAR POSICION DE SUS 0. COMPROBAR TEOREMA 3.4.

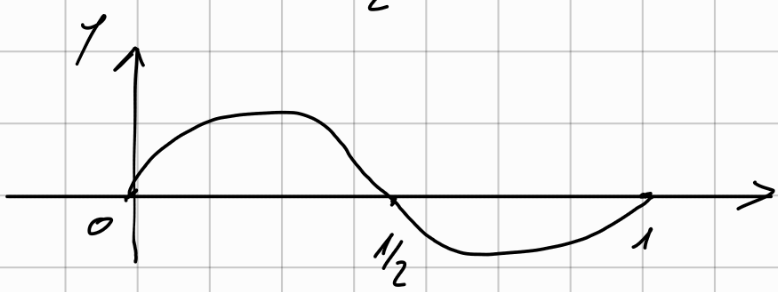
$n = 1$ $\quad y(x) \propto e^{-x} \sin(\pi x)$

$y(x) = 0$ ONLY AT $x = 0, 1$



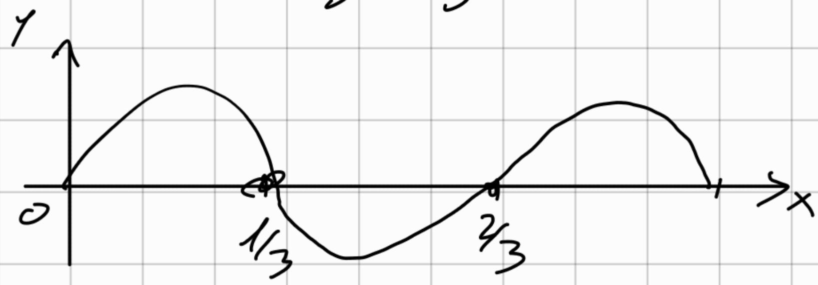
$n = 2$ $\quad y(x) = e^{-x} \sin(2\pi x)$

$y(x) = 0$ AT $x = 0, \frac{1}{2}, \pi$



$m=3$ $\quad y(x)=e^{-x}\sin(3\pi x)$

$y(x)=0$ AT $x=0, \frac{1}{3}, \frac{2}{3}, x=1$



COMPROBACIÓN DEL TEOREMA 3.4

SI p Y r SON POSITIVAS EN $[a,b]$ LOS AUTOVALORES DEL PROBL. DE ST-L.

$$(p\varphi')'+\lambda r\varphi=0 \qquad \varphi(a)=\varphi(b)=0$$

SON SIMPLES Y SE PUEDEN ORDENAR DISCRETAMENTE

$$0<\lambda_1<\lambda_2<\dots$$

Y $\lambda_m \Rightarrow$ AUTOFUNCIÓN CON $m-1$ CEROS EN (a,b)

AQUÍ $p=r=e^{2x}$ ✓

TODO EL RESTO ES CIERTO

(c) COMPROBAR QUE EL TEOREMA NO ADMITE AUTOVALORES COMPLEJOS (NO REALES)

YA CALCULAMOS AUTOVALORES Y SON REALES

ES CONSECUENCIA DE QUE $\mathcal{L}$ ES HERMÍTICO Y $\Omega > 0$ EN $[a, b]$.

DÍA 2

7 Considerar problema St.-L.

$$(x^2 y')' = -\lambda y \quad \text{en } [1,2]$$

$$y(1) = y(2) = 0$$

a Demonstrar que autovalores y autofunciones

$$\lambda_n = \frac{1}{2} + \frac{(n\pi)^2}{(\ln 2)^2} \qquad y_n = \frac{1}{\sqrt{x}} \sin\left(\frac{n\pi \ln x}{\ln 2}\right)$$

Solución ①: verifica directamente ecuación.

Solución ②:

Del probl. ⑥ sabemos que

$$\frac{d}{d\tilde{x}}\left(e^{2\tilde{x}} \frac{d\tilde{y}}{d\tilde{x}}\right) + \tilde{\lambda} e^{2\tilde{x}} \tilde{y}(\tilde{x}) = 0$$

$$e^{2\tilde{x}}\left[\tilde{y}'' + 2\tilde{y}' + \tilde{\lambda}\tilde{y}\right] = 0 \qquad \tilde{y}(0)=0 \quad \tilde{y}(1)=0$$

$$\Downarrow$$

$$\tilde{\lambda}_n = 1 + (n\pi)^2, \qquad \tilde{y}_n(\tilde{x}) = c\, e^{-\tilde{x}} \sin(n\pi \tilde{x})$$

Pongamos $\boxed{\tilde{x} = \frac{\ln x}{\ln 2} = \log_2 x} \Rightarrow \boxed{2^{\tilde{x}} = x = e^{\tilde{x}\ln 2}}$

$$\boxed{\begin{array}{lll} x = 1 & \Rightarrow & \tilde{x} = 0 \\ x = 2 & \Rightarrow & \tilde{x} = 1 \end{array}}$$

$$\boxed{a^{b\tilde{x}} = e^{b\tilde{x}\ln a}} \qquad \boxed{\frac{d}{d\tilde{x}} = b \ln a\, e^{b\tilde{x}\ln a} = b \ln a\, a^{b\tilde{x}}}$$

$$e^{-\tilde{x}} = e^{-\left(\frac{\ln x}{\ln 2}\right)} = (x)^{-1/\ln 2}$$

$$\tilde{Y}_n(\tilde{x}(x)) = c\, x^{-1/\ln 2} \sin\left(\frac{n\pi \ln x}{\ln 2}\right)$$

RELACIÓN ENTRE $\tilde{Y}_n$ NO NOTA Y Y_n NOTA

$$= \alpha(x)\, y_n(x) = \frac{\alpha(x)}{x^{1/2}} \sin\left(\frac{n\pi \ln x}{\ln 2}\right)$$

$$\Longrightarrow \boxed{\alpha(x) = x^{\frac{1}{2} - \frac{1}{\ln 2}} = 2^{\frac{\tilde{x}}{2}} e^{-\tilde{x}} \overset{!}{=} e^{\tilde{x}\left(\frac{1}{2}\ln 2 - 1\right)}}$$

$$\frac{d}{dx} = \frac{d\tilde{x}}{dx}\frac{d}{d\tilde{x}} = \frac{1}{x \ln 2}\frac{d}{d\tilde{x}} = \frac{1}{2^{\tilde{x}}\ln 2}\frac{d}{d\tilde{x}}$$

AHORA

$$(x^2 y')' + \lambda y =$$

$$= \frac{1}{2^{\tilde{x}}\ln 2}\frac{d}{d\tilde{x}}\left(x^2 \frac{1}{x\ln 2}\frac{d}{d\tilde{x}}\left(\frac{1}{\alpha}\tilde{y}\right)\right) + \lambda\frac{\tilde{y}}{\alpha}$$

$$= e^{-\tilde{x}\ln 2}\frac{1}{(\ln 2)^2}\frac{d}{d\tilde{x}}\left(e^{\tilde{x}\ln 2}\frac{d}{d\tilde{x}}\left(e^{(1-\frac{1}{2}\ln 2)\tilde{x}}\,\tilde{y}\right)\right) + \lambda\, e^{(1-\frac{1}{2}\ln 2)\tilde{x}}\,\tilde{y}$$

$$= \frac{e^{-\tilde{x}\ln 2}}{(\ln 2)^2}\frac{d}{d\tilde{x}}\left(e^{\tilde{x}\ln 2}\left((1-\tfrac{1}{2}\ln 2)\,\tilde{y} + \tilde{y}'\right)e^{(1-\frac{1}{2}\ln 2)\tilde{x}}\right) + \lambda\, e^{(1-\frac{1}{2}\ln 2)\tilde{x}}\,\tilde{y}$$

$$= \frac{e^{-\tilde{x} h_2}}{(h_2)^2} \frac{d}{d\tilde{x}} \left(e^{(1+\frac{1}{2}h_2)\tilde{x}} \left((1-\tfrac{1}{2}h_2)\tilde{y} + \tilde{y}' \right) \right) + \lambda e^{(1-\frac{1}{2}h_2)\tilde{x}} \tilde{y}$$

$$= \frac{e^{-\tilde{x} h_2}}{(h_2)^2} \left[(1+\tfrac{1}{2}h_2)\left((1-\tfrac{1}{2}h_2)\tilde{y} + \tilde{y}'\right) + (1-\tfrac{1}{2}h_2)\tilde{y}' + \tilde{y}'' \right] e^{(1+\frac{1}{2}h_2)\tilde{x}}$$

$$+ \lambda e^{(1-\frac{1}{2}h_2)\tilde{x}} \tilde{y}$$

$$= \frac{e^{(1-\frac{1}{2}h_2)\tilde{x}}}{(h_2)^2} \left[\tilde{y}'' + 2\tilde{y}' + \left(1 - \tfrac{1}{4}(h_2)^2\right)\tilde{y} \right] + \lambda e^{(1-\frac{1}{2}h_2)\tilde{x}} \tilde{y}$$

$$= \frac{e^{(1-\frac{1}{2}h_2)\tilde{x}}}{(h_2)^2} \left[\tilde{y}'' + 2\tilde{y}' + \underbrace{\left((h_2)^2\lambda + 1 - \tfrac{1}{4}(h_2)^2\right)}_{\tilde{\lambda}} \tilde{y} \right]$$

$$\boxed{\begin{aligned} \lambda &= \frac{1}{(h_2)^2}\left(\tilde{\lambda} - 1 + \tfrac{1}{4}(h_2)^2\right) \\ &= \frac{1}{4} + \frac{\tilde{\lambda} - 1}{(h_2)^2} \end{aligned}}$$

$$\lambda_n = \frac{1}{4} + \frac{\cancel{1} + (n\pi)^2 - \cancel{1}}{(h_2)^2} = \frac{1}{4} + \frac{(n\pi)^2}{(h_2)^2} \quad ✓$$

b) SE CUMPLE T. 3.4 ✓

PROP. ORTOGONALIDAD.

$$(y_n, y_p)_r = \int_1^2 dx \underset{\overset{\parallel}{r}}{1} \frac{1}{x} \sin\left(\frac{n\pi \ln x}{\ln 2}\right) \sin\left(\frac{m\pi \ln x}{\ln 2}\right)$$

$$d\tilde{x} = \frac{dx}{x} \frac{1}{\ln 2}$$

$$= \ln 2 \int_0^1 d\tilde{x} \sin(n\pi \tilde{x}) \sin(m\pi \tilde{x})$$

$$= 0 \quad \text{SI} \quad n \neq m$$

8 a DETERMINAR VALORES DE α PARA LOS QUE

$$\frac{d^2y}{dx^2} + \lambda y(x) = 0 \,, \qquad \alpha\, y(0) + y'(0) = y'(1) = 0$$

EN $[0,1]$ TIENE AUTOVALORES <u>NEGATIVOS</u>

EC. HOMOGÉNEA CON COEF. CONSTANTES $\Rightarrow$

$$y = e^{mx}$$

$$\Rightarrow e^{mx}(m^2+\lambda) = 0 \Rightarrow m_{\pm} = \pm\sqrt{-\lambda}$$

DISTINTOS PORQUE $\lambda < 0$

$$\boxed{y = C_+ e^{x\sqrt{-\lambda}} + C_- e^{-x\sqrt{-\lambda}}}$$

$$\boxed{y' = \sqrt{-\lambda}\left(C_+ e^{x\sqrt{-\lambda}} - C_- e^{-x\sqrt{-\lambda}}\right)}$$

CONDICIONES DE CONTORNO:

$$y'(1) = \sqrt{-\lambda}\left(C_+ e^{\sqrt{-\lambda}} - C_- e^{-\sqrt{-\lambda}}\right) = 0$$

$$\Rightarrow \boxed{C_- = C_+ e^{2\sqrt{-\lambda}}}$$

EN 0

$$\hookrightarrow \alpha(C_+ + C_-) + \sqrt{-\lambda}(C_+ - C_-)$$

$$= C_+\left[\alpha\left(1 + e^{2\sqrt{-\lambda}}\right) + \sqrt{-\lambda}\left(1 - e^{2\sqrt{-\lambda}}\right)\right] = 0$$

$$C_+ \neq 0 \Rightarrow \alpha = -\sqrt{-\lambda}\,\frac{1-e^{2\sqrt{-\lambda}}}{1+e^{2\sqrt{-\lambda}}}$$

$$= \sqrt{-\lambda}\,\frac{\sinh\sqrt{-\lambda}}{\cosh\sqrt{-\lambda}}$$

$$= \sqrt{-\lambda}\,\tanh\sqrt{-\lambda}$$

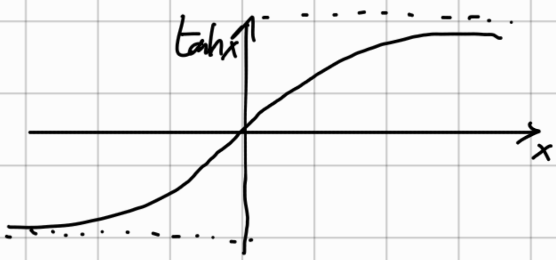
ALTERNATIVAMENTE

$$\begin{array}{r l} y'(1) & \left\{ \begin{array}{l} e^{\sqrt{-\lambda}}C_+ - e^{-\sqrt{-\lambda}}C_- = 0 \\ (\alpha+\sqrt{-\lambda})C_+ + (\alpha-\sqrt{-\lambda})C_- = 0 \end{array}\right. \\ \alpha y(0) + y'(0) & \end{array}$$

LINEARM. DEP. $\Rightarrow$ $\det = e^{\sqrt{-\lambda}}(\alpha-\sqrt{-\lambda}) + e^{-\sqrt{-\lambda}}(\alpha+\sqrt{-\lambda}) = 0$

$$\boxed{\alpha = \sqrt{-\lambda}\,\tanh\sqrt{-\lambda}}$$

Si $\lambda < 0 \Rightarrow \sqrt{-\lambda} \in \mathbb{R}_{>0}$



$\Rightarrow \boxed{\alpha > 0}$

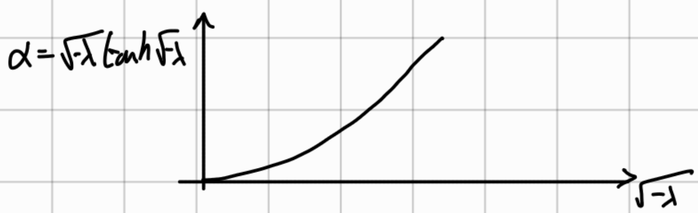
b) ESTUDIA LOS DIFERENTES CASOS.

$\lambda < 0$

$$y(x) = C_+ \left(e^{\sqrt{-\lambda}\, x} + e^{2\sqrt{-\lambda}} e^{-\sqrt{-\lambda}\, x} \right)$$
$$= C_+ e^{\sqrt{-\lambda}} \left(e^{\sqrt{-\lambda}(x-1)} + e^{-\sqrt{-\lambda}(x-1)} \right)$$
$$= 2C_+ e^{\sqrt{-\lambda}} \cosh\left(\sqrt{-\lambda}\,(x-1)\right)$$

$$\boxed{y(x) = A \cosh\left(\sqrt{-\lambda}\,(x-1)\right)}$$

DADO α, QUEREMOS SABER EL AUTOVALOR λ.
DEBERIAMOS RESOLVER
$$\alpha = \sqrt{-\lambda}\, \tanh \sqrt{-\lambda}$$
QUE ES UNA EC. TRANSCENDENTAL.
PERO PODEMOS DECIR QUE CONOCIENDO EL GRÁFICO



EXISTE UN ÚNICO VALOR DE λ POR CADA α.

$\lambda = 0$

$$\frac{d^2 y}{dx^2} = 0 \Rightarrow \boxed{y = C_0 + C_1 x}$$
$$\boxed{y' = C_1}$$

$$y'(1) = 0 \Rightarrow C_1 = 0$$

$$\alpha\, y(0) + y'(1) = \alpha C_0 = 0$$

$\alpha \neq 0$ — $C_0 \Rightarrow$ TRIVIAL

$\alpha = 0$ — $\forall C_0 \Rightarrow \boxed{y = C_0}$

$\lambda > 0$ $\quad \sqrt{-\lambda} = i\sqrt{\lambda}$

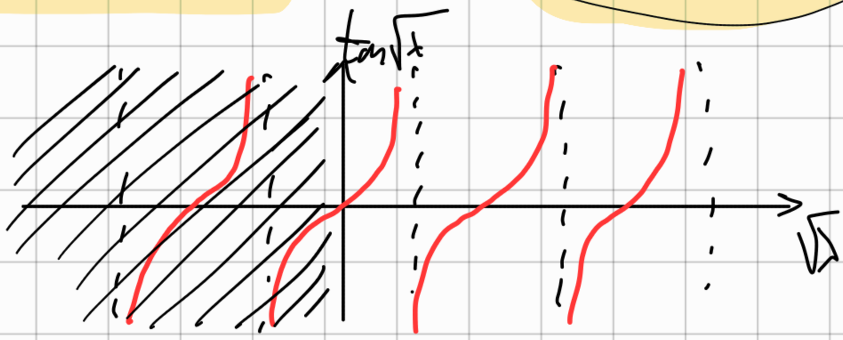
$$\Rightarrow \quad y(x) = A\cos\left(\sqrt{\lambda}(x-1)\right)$$

$$\alpha = i\sqrt{\lambda}\tanh(i\sqrt{\lambda}) = -\sqrt{\lambda}\tan\sqrt{\lambda}$$

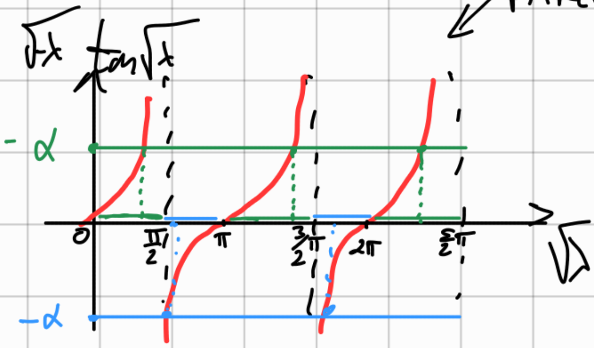
DADO α, SABEMOS ENCONTRAR λ?

RECORDAMOS QUE

$\sqrt{\lambda} > 0$



PARECIDO



$\Rightarrow$ CUANDO RESOLVEMOS POR λ TENEMOS UNA SOLUCIÓN EN CADA INTERVALO $\left(-\frac{\pi}{2} + \pi n, \frac{\pi}{2} + \pi n\right)$ $n \in \mathbb{Z}$

+ CONSTRAINT $\sqrt{\lambda} > 0$

$$\alpha < 0 \Rightarrow \sqrt{\lambda} \in \left(\pi n, \pi\left(\tfrac{1}{2} + n\right)\right) \quad n \in \mathbb{N}$$

THIS FIRST $\rightarrow$ $\alpha = 0 \Rightarrow \sqrt{\lambda} = \{\pi(n+1)\} \quad n \in \mathbb{N}$

$$\alpha > 0 \Rightarrow \sqrt{\lambda} \in \left(\pi\left(\tfrac{1}{2} + n\right), \pi(1+n)\right) \quad n \in \mathbb{N}$$

N.B.: SIEMPRE EXISTE UN $\lambda_{min.}$ PORQUE $\sqrt{\lambda} > 0$

9 ENCUENTRA AUTOVALORES Y AUTOFUNCIONES DE

$$y'' + 2y' + (1-\lambda)y = 0 \qquad 0 \leq x \leq \pi$$

CON $y'(0) = 0 = y'(\pi)$. VERIFICA ORTOGONALIDAD PARA $\lambda \neq 0$

$$y = e^{mx}$$

$$e^{mx}\left(m^2 + 2m + (1-\lambda)\right) = 0$$

$$m_\pm = -1 \pm \sqrt{1-(1-\lambda)} = -1 \pm \sqrt{\lambda}$$

$$y = C_+ e^{(-1+\sqrt{\lambda})x} + C_- e^{(-1-\sqrt{\lambda})x}$$

$$y' = C_+(-1+\sqrt{\lambda})e^{(-1+\sqrt{\lambda})x} + C_-(-1-\sqrt{\lambda})e^{(-1-\sqrt{\lambda})x}$$

$$\begin{cases} y'(0): & (-1+\sqrt{\lambda})C_+ + (-1-\sqrt{\lambda})C_- = 0 \\ y'(\pi): & (-1+\sqrt{\lambda})e^{(-1+\sqrt{\lambda})\pi}C_+ + (-1-\sqrt{\lambda})e^{(-1-\sqrt{\lambda})\pi}C_- = 0 \end{cases}$$

$$\det = (1-\lambda)e^{(-1-\sqrt{\lambda})\pi} - (1-\lambda)e^{(-1+\sqrt{\lambda})\pi}$$

$$= (1-\lambda)e^{-\pi}\left(e^{-\sqrt{\lambda}\pi} - e^{\sqrt{\lambda}\pi}\right) = 0$$

I $\lambda = 1 \Rightarrow m_+ = 0, \quad m_- = -2$

$C_- = 0$

$$y = C_+ = \text{const.}$$

II $$e^{-\sqrt{\lambda}\pi} - e^{\sqrt{\lambda}\pi} = 0$$

$$e^{2\sqrt{\lambda}\pi} = 1$$

$$2\sqrt{\lambda}\pi = 2\pi i n$$

$$\sqrt{\lambda} = in \qquad n \in \mathbb{Z}$$

$$\Rightarrow \lambda < 0 \qquad \lambda = -n^2$$

$$(-1+\sqrt{\lambda})C_+ + (-1-\sqrt{\lambda})C_- = 0$$

$$C_- = C_+ \frac{\sqrt{\lambda}-1}{\sqrt{\lambda}+1} = C_+ \frac{in-1}{in+1}$$

$$m_\pm = -1 \pm in$$

$$y = C_+ e^{(-1+\sqrt{\lambda})x} + C_- e^{(-1-\sqrt{\lambda})x}$$

$$= C_+ \left[e^{(-1+in)x} + \frac{in-1}{in+1} e^{(-1-in)x} \right]$$

$$= \frac{C_+}{in+1} e^{-x} \left[(in+1)e^{inx} + (in-1)e^{-inx} \right]$$

$$= A\, e^{-x}\left[n\cos(nx) + \sin(nx) \right]$$

ORTOGONALIDAD

EN ESTE CASO

$$\frac{p'}{p} = 2 \qquad \ln p = 2x \qquad p = e^{2x}$$

$$(1-\lambda)y \Rightarrow \quad -1 = \frac{\lambda r}{p}$$

$$\Rightarrow \quad r = -p = -e^{2x}$$

$$\int_0^{\pi} dx\, |r(x)|\, y_n(x)\, y_m(x)$$

$$= \int_0^{\pi} dx\, e^{2x} e^{-2x} \left[n\cos(nx) + \sin(nx) \right]\left[m\cos(mx) + \sin(mx) \right]$$

$$= \int_0^{\pi} \frac{dx}{4} \Big[e^{i(n+m)x}(nm - in - im - 1)$$

$$+ e^{i(n-m)x}(nm + in - im + 1)$$

$$+ e^{i(-n+m)x}(nm - in + im + 1)$$

$$+ e^{i(-n-m)x}\left(nm + in + im - 1 \right)\Big]$$

$$= \frac{1}{4}\Bigg[\frac{(nm - in - im - 1)}{n+m}\left(e^{i(n+m)\pi} - 1\right)$$

$$+ \frac{nm + in - im + 1}{n-m}\left(e^{i(n-m)\pi} - 1\right)$$

$$+ \frac{nm - in + im + 1}{-n+m}\left(e^{i(-n+m)\pi} - 1\right)$$

$$+ \frac{nm + in + im - 1}{-n-m}\left(e^{-i(n+m)\pi} - 1\right)$$

$n+m = s \qquad n-m = d$

$n = \frac{s+d}{2} \qquad m = \frac{s-d}{2}$

$nm = \frac{s^2-d^2}{4}$

$e^{is\pi} = e^{-is\pi}$

$$\left(\frac{s^2-d^2}{4} - is - 1\right)\frac{1}{s}\left(e^{is\pi} - 1\right)$$

$$- \left(\frac{s^2-d^2}{4} + is - 1\right)\frac{1}{s}\left(e^{-is\pi} - 1\right)$$

$$= \frac{1}{s}\left(e^{-is\pi} - 1\right)\left(-2is\right) = -2i\left(e^{is\pi} - 1\right)$$

$$\frac{\frac{s^2}{4} - d^2 + id + 1}{d}\left(e^{id\pi} - 1\right)$$

$$- \frac{\frac{s^2 - d^2}{4} - id + 1}{d}\left(e^{-id\pi} - 1\right)$$

$$= \left(e^{id\pi} - 1\right)\frac{1}{d}\, 2id = 2i\left(e^{id\pi} - 1\right)$$

SI m, n AMBOS PARES O IMPARES

$\Rightarrow$ s, d PARES $\Rightarrow \left(e^{is\pi} - 1\right) = \left(e^{id\pi} - 1\right) = 0$

SI m PAR Y n IMPAR (O VICEVERSA)

$\Rightarrow$ s, d IMPARES $\Rightarrow \left(e^{is\pi} - 1\right) = \left(e^{id\pi} - 1\right) = -2$

$$-2i(-2) + 2i(-2) = 0 \checkmark$$

EXCEPCIÓN: SI $n = m \Rightarrow d = 0$

$\Rightarrow$ NO SE PUEDE DIVIDIR POR $\frac{1}{n-m}$